

● HANDPICKED FREE RESOURCES - 2026

The 5 VLSI Foundations

The five things to learn first - and the best free course for each. Digital Electronics, Verilog, CMOS, Static Timing Analysis, and Python/TCL. All links are free and clickable.

- 01 **Digital Electronics**
- 02 **Verilog**
- 03 **CMOS Basics**
- 04 **Static Timing Analysis**
- 05 **Python / TCL Scripting**

Start here

Master these in order. Each entry is the single best free option.

01

Digital Electronics

NESO ACADEMY

Neso Academy - Digital Electronics (YouTube)

<https://www.youtube.com/playlist?list=PLBlnK6fEygRjMH3mWf6kwqiTbT798eA0m>

The foundation of everything in VLSI. Number systems, Boolean algebra, K-maps, combinational and sequential logic. Pair with Morris Mano's textbook for depth.

02

Verilog

NPTEL + HDLBITS

NPTEL - Hardware Modeling using Verilog (IIT Kharagpur)

https://onlinecourses.nptel.ac.in/noc23_cs76/preview

HDLBits - hands-on practice (do this alongside)

<https://hdlbits.01xz.net/>

Watch to learn the language; use HDLBits to actually get fluent. Verilog doesn't stick without writing it.

03

CMOS Basics

NPTEL

NPTEL - CMOS Digital VLSI Design (IIT Roorkee)

https://onlinecourses.nptel.ac.in/noc21_ee09/preview

How digital logic actually works inside the chip - MOS transistor, CMOS inverter, logic design, power, SPICE. This is the digital-focused course (the right pick).

STA: the one to watch

One of the most important concepts in real chip design. The best free video course - with the exact lectures that cover STA.

VLSI Physical Design with Timing Analysis

NPTEL - IIT Roorkee - Prof. Bishnu Prasad Das

YouTube playlist:

https://www.youtube.com/playlist?list=PLLy_2iUCG87Bny6CcGkCanv1HuXwr4-_W

NPTEL course page:

<https://nptel.ac.in/courses/108107380>

LECTURES THAT COVER STA

Week 2 (STA-1): Lec 6-11 Week 3 (STA-2): Lec 12-17

Lec 6-11: intro, timing arcs, setup/hold, flip-flop & latch STA

Lec 12-17: time borrowing, OCV/CRPR, MMMC, statistical STA

Plus OpenSTA tool demo around Lec 56

THE ONE TO WATCH

Note: Synopsys PrimeTime / Cadence Tempus have NO free video course - vendor training is license-gated, and PrimeTime/Tempus courses online are paid third-party. Use OpenSTA free instead.

Scripting

Python for automation, TCL for the EDA tools. Pick a Python course by your style; the TCL one is VLSI-specific.

Python

FULL COURSE

Shradha Khapra - Python Language Full Course (2026)

<https://youtube.com/playlist?list=PLGjplNEQ1it8-0CmoljS5yeV-GlKSUEt0>

9-video full course, updated for 2026 - best fit for an Indian engineering audience.

TCL Scripting

EDA-SPECIFIC

Chip Design with Rashid - Learn EDA tools language: TCL Scripting

<https://youtube.com/playlist?list=PL0-xus8sJBCRpVtxjGLwB12YyQKQH2wRW>

TCL framed for actual EDA tool flows - not generic TCL. The relevant one for chip work.

Learn in order: Digital -> Verilog -> CMOS -> STA -> Scripting. Watch less, build more - every topic here is one you should practice, not just play.